AI DRIVEN KINECTIC TILES

A DESIGN PROJECT-1 (5CS1990) REPORT

Submitted by

THUNUGUNTLA RAHUL SANTOSH (2023BCSE07AED274)

P PRATHYUSH KUMAR (2023BCSE07AED246)

S JESWIN (2023BCSE07AED245)

KRISHNA HRUDAY (2023BCSE07AED275)

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In

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Under the Supervision of

Dr. Lopa Mandal

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ALLIANCE SCHOOL OF ADVANCED COMPUTING

ALLIANCE UNIVERSITY, BENGALURU

DEPARTMENT OF CSE(AIML)

CERTIFICATE

This is to certify that the Design Project -1 work entitled "AI-Driven Kinetic Tiles" is the bonafide work done by THUNUGUNTLA RAHUL SANTOSH (2023BCSE07AED274), P PRATHYUSH KUMAR (2023BCSE07AED246), S JESWIN (2023BCSE07AED245), and KRISHNA HRUDAY (2023BCSE07AED275) submitted in partial fulfilment of the requirements for the award of the degree Bachelor of Technology in COMPUTER SCIENCE AND ENGINEERING (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING) during the year 2025–2026.

Supervisor

Assistant Professor/ Associate Professor /Professor

Dept. of CSE

ALLIANCE UNIVERSITY

ALLIANCE SCHOOL OF ADVANCED COMPUTING

DECLARATION

This is to declare that the report titled **"AI-Driven Kinetic Tiles"** has been made for the partial fulfilment of the Programme – Bachelor of Technology in **Computer Science and Engineering (AIML)**, under the guidance of Dr. Lopa Mandal. I confirm that this report truly represents my work undertaken as a part of **my Design Project -1** work. This work is not a replication of work done previously by any other person. I also confirm that the contents of the report and the views contained therein have been discussed and deliberated with the Mentor.

NAME	REGESTRATION NO.	SIGNATURE
THUNUGUNTLA RAHUL SANTOSH	2023BCSE07AED274	
P PRATHYUSH KUMAR	2023BCSE07AED246	
S JESWIN	2023BCSE07AED245	
KRISHNA HRUDAY	2023BCSE07AED275	

ALLIANCE UNIVERSITY

ALLIANCE SCHOOL OF ADVANCED COMPUTING

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THUNUGUNTLA RAHUL SANTOSH (2023BCSE07AED274)

S JESWIN (2023BCSE07AED245)

KRISHNA HRUDAY (2023BCSE07AED275)

P PRATHYUSH KUMAR (2023BCSE07AED246)

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1. INTRODUCTION

With cities expanding rapidly and global energy demands on the rise, the need for sustainable and smart energy solutions is becoming increasingly important. Our project, **AI-Driven Kinetic Tiles**, introduces a creative approach that combines renewable energy generation with intelligent data analysis. These tiles capture the energy produced by people walking or vehicles moving over them and convert it into electricity.

By using artificial intelligence, the system becomes more efficient and adaptive. It studies real-time information such as foot traffic, energy generation, and usage patterns to manage energy storage and distribution effectively. Through predictive analysis, it ensures that the available energy is used in the best possible way, supporting both performance and sustainability.

This technology can play a key role in the development of smart cities, especially in areas like transportation hubs, campuses, and other busy public spaces. Besides generating clean energy, it can also help with crowd monitoring, infrastructure management, and better urban planning.

The main aim of this project is to design a reliable system that shows how **AI-powered kinetic tiles** can support energy-efficient, data-driven, and eco-friendly infrastructure for the cities of the future.

2. LITERATURE REVIEW

Energy-harvesting floor tiles have gained significant attention as a sustainable means to address the rising energy requirements of IoT devices, smart buildings, and low-power electronics. These systems capture mechanical energy from human movements or vehicle loads, transforming it into electrical power, while also providing additional benefits like sensing and monitoring capabilities. This section presents an overview of notable studies in this field, highlighting their goals, scope, and limitations.

The energy-harvesting floor tiles have been of great interest in developing a sustainable solution to tackle the increasing energy demand of IoT devices, smart buildings, and low-power electronics. These systems convert mechanical energy from human movements or vehicle loads into electrical power, while also providing other additional benefits such as sensing and monitoring capabilities. This section gives an overview of the prominent studies in the area, including their objectives, scope, and limitations.

1. **Piezoelectric Energy Harvesting Floor Mat for Low Power Applications [1]**

The aim of this work was to demonstrate the feasibility of harvesting human kinetic energy from footsteps by using a simple, low-cost piezoelectric mat. The primary goal was to show the possibility of converting small actions like walking into usable electricity for everyday low-power applications.

SCOPE: The researchers constructed a prototype with 35 piezoelectric discs in a series-parallel arrangement and mounted them on a
300×300 mm wooden tile. The harvested energy was directed to simple loads such as LEDs and small rechargeable batteries, validating the concept for basic applications. The focus was to be low-cost, easy to make, and a proof-of-concept rather than for long-time and large-scale uses.

Limitation:
• Generates only microvolt-level outputs: The energy harvested is too small to be useful for practical devices beyond LEDs.
• It can power only LEDs or very small loads: The system is limited to demonstration-level applications, not real-world energy needs.
• Fragile ceramic disks that easily break: The piezoelectric disks crack or wear out quickly under repeated foot traffic.
• Not scalable for public use: The design is incapable of supporting heavy, continuous usage in public spaces like walkways or stations.

2. Response Surface Methodology Analysis of Energy Harvesting System over Pathway Tiles [2]

Objective: This paper targets enhancing the performance of piezoelectric floor tiles by employing statistical optimization. The authors used Response Surface Methodology to model and analyze how weight load, the number of piezoelectric sensors, and wiring configurations may affect the harvested energy.

Scope: Tiles embedded with 40 mm PZT discs were subjected to different loads ranging from 50–150 kg. The RSM model was compared against the experimental results in showing how predictive analysis could reduce trial-and-error in tile design. The study highlighted how systematic parameter optimization can aid in developing more efficient piezoelectric energy harvesting systems for walkways and pathways.

Limitation:
• Predicted efficiency was higher than actual results: Simulations displayed good performance, while experiments resulted in much less power.
• Experimental power output remained low: Even optimized tiles did not produce enough energy for meaningful applications.
• Tested Only in Lab Conditions: No trials were done on real pedestrian pathways, thus field performance is unknown.
• High material and circuit costs: Using many sensors and complex circuits increases the cost, thus limiting real-world applications.

3. Triboelectric Energy-Harvesting Floor Tile [3]

Objective: To investigate the TENGs as a feasible alternative to piezoelectric tiles with a view to cost reduction, and the simplification of fabrication. In the presented research work, the influence of design factors such as material thickness, gap width, and stepping frequency on triboelectric tile performance was reviewed.

Scope: A tile with a contact–separation mode was fabricated using PTFE and aluminum as triboelectric layers. Tests showed that with a 0.2 mm thick layer of PTFE, an energy of ~7.69 mJ per step was produced, which can light up 100–150 LEDs at the same time. This work compared various thicknesses and operational frequencies while determining the conditions for maximum triboelectric efficiency.

Limitation:
• Energy per step was still low: Although cheaper, the energy harvested was less than optimized piezoelectric tiles.
• Less efficient overall: The triboelectric conversion efficiency is lower, reducing viability for continuous energy harvesting.
• Durability untested: The PTFE and aluminum layers could degrade or lose efficiency under thousands of footsteps.
• Validated only in controlled setups - Real-world pedestrian or outdoor testing was not done.

4. Development of a Pavement-Embedded Piezoelectric Harvester in a Real Traffic Environment [4]

Objective: To assess the real-world feasibility of piezoelectric energy harvesting by embedding harvesters directly into road pavement. Other than controlled lab-based studies, the project seeks to confirm the durability and performance in actual traffic loads.

Scope: The system was installed in a Brazilian toll plaza with heavy vehicle traffic that provided essentially constant load conditions. Energy harvested was stored and supplied successfully to roadway applications like cat's-eye reflectors, demonstrating the feasibility of piezoelectric harvesters in transportation infrastructure. This was one of the first large-scale outdoor piezoelectric energy harvesting demonstrations under realistic operating conditions.

Limitation:
• Limited to one site and short-term trials; results are not generalizable without broader, longer-term testing.
• Durability under environment untested: Factors like rain, heat, and continuous
Traffic stress was not studied in any detail.
• High installation and maintenance costs are ignored. In-road embedding of piezoelectric harvesters may not be economically feasible.

5. Self-Powered and Self-Sensing Floor Tiles and Fall Detection System [5]

Objective: The following study aimed to incorporate energy harvesting and real-time sensing in one system for smart floor tiles, which would be self-powered and could detect falls without using any external sensors. The goal was to reduce dependency on batteries while allowing health-monitoring applications.

Scope: The developed prototype consisted of a 28 tile array, with each tile implemented using piezoelectric buzzers for energy harvesting, an energy management circuit, and a BLE SoC. Additionally, every step or fall generates enough energy to transmit multiple BLE packets. The CNN was used to process the data collected from the tile array and reported 90% fall detection accuracy with 100% identification of normal walking during controlled tests.

Limitation:
• Tested with small groups of users and only short-term: Accuracy was demonstrated but may not generalize to larger populations.
• BLE communication only: it has only enough energy harvested for Bluetooth packets, and not heavier IoT applications.
• Algorithm dependency: The accuracy of fall detection (90%) is highly dependent on the training of CNN with small datasets; scalability to multiple users, noise, or simultaneous events remains unproven.

6. Low-to-moderate Vibration Sensible Cost-effective Electricity Generating Floor Tile [6]

Objective: The objective of this research was to design a cost-effective electromagnetic floor tile capable of converting human tread into an electrical energy output. Particular attention was given to the use of low-cost materials and ease of construction so that the design could be reproduced for demonstrations, small-scale energy harvesting projects, or in areas where expensive commercial solutions like Pavegen are not viable.

Scope: The design targets low-power, small-scale applications for driving LEDs, mobile charging hubs, and basic sensors in open public places like bus stations, corridors, or schools. It is also meant for academic demonstrations in order to explain the principle of electromagnetic induction for renewable energy.

Limitation:
• Output power of the system highly relies on the weight and force of the user; this makes the energy output inconsistent.
• Harvested power is much lower than in commercial solutions like Pavegen, capable of several watts per step.
• Frictional losses as well as imperfect centering of magnets and coils lead to energy loss.

7. Energy Harvesting through Floor Tiles [7]

Objective: The purpose was to use piezoelectric discs inlaid in floor tiles to transform human footsteps into electrical energy, which will be accumulated in batteries for later usage.

Scope: The scope of application for these would principally be low-power electronics, such as wireless IoT devices, environmental sensors, mobile charging docks, and LED lights. It can also be in buildings for smart monitoring systems or demonstrations for energy-saving purposes.

Limitations:
• It cannot power medium or high-load devices.
• Needs continuous foot traffic to build up its store of energy; otherwise, storage is very slow.
• Piezo discs' durability in everyday use hasn't been tested.
• Environmental Sensitivity: Exposure to moisture, dust, and vibration reduces life span and efficiency.

8. Design and Fabrication of in Touch Power Generation Tiles [8]

Objective: The objective was to design scalable and eco-friendly "inTouch" tiles using piezoelectric sensors for integration into urban infrastructure to support smart city initiatives.

Scope: The target scope includes high-footfall areas like railway stations, religious sites, pavements, stadiums, and smart buildings where collective energy from large crowds can be harvested to power streetlights, smart sensors, and IoT-based city services.

Limitations:
• Long-term durability and maintenance costs are unknown.
• Not fit for heavy and continuous loads.
• Application-wise, it requires integration with smart grid systems, making its practicality more cumbersome.

9. Development of Highly Efficient Hybrid Kinetic-Solar Energy Harvesting System [9]

Objective: Putting kinetic piezoelectric harvesting together with solar photovoltaic cells will create a hybrid energy system that continuously and reliably delivers power to the end-users.

Scope: The scope includes IoT devices, wireless sensors, and other small remote systems that require a constant energy supply without depending only on footsteps. It comes in handy for places that have sunlight and an intermittent flow of foot traffic.

Limitations:
• The hybrid system is complex compared to standalone systems; thus, careful integration of solar panels and piezoelectric modules is required.
• Manufacturing and installation costs are higher.
• Requires both footsteps and sunlight to function best.

10. Development of Piezoelectric Tiles for 48 Wh Energy (16W LED for 3 hrs) [10]

Objectives: The aim is to design and fabricate a piezoelectric tile for generating and storing 48 Wh of energy, which can power a 16W LED bulb for three hours.

Scope: Designed for educational demonstrations, small-scale renewable applications, and public buildings with sufficient student or visitor footfall to generate usable power.

Limitations:
• Not optimized for durability; the discs may wear out under long-term usage.
• More suitable for demonstration and education rather than real-world urban deployment.
• Requires a large installation area to make the system practical.

11. Sustainable tiles for renewable energy harvesting using integrated solar PV, thermoelectric generator, and piezoelectric technologies [11]

Objectives:
• To design a hybrid tile capable of harvesting solar, thermal, and mechanical energy simultaneously.
•To replace a part of natural sand with torched fly ash for better sustainability and resource efficiency.
• To integrate PV, TEG, and piezoelectric systems into a single modular tile design

Scope:
• It can be deployed in Urban Walkways, Parks, Pedestrian Zones, and building exteriors.
• Applicable in smart cities that require sustainable power for IoT gadgets, lighting, and environmental monitoring.
• Provides a multi-source harvesting approach, reducing dependency on one single renewable source.

Limitations:
• Prototype tested only in controlled environments and limited field trials.
• Long-term durability and maintenance under outdoor exposure remain untested.
• Energy output is modest (few watts per m^2) and is not capable of large-scale independent use.
• Integrating into existing pavement and retrofitting can be difficult.

12. Human Footsteps-based Energy Harvesting Using Piezoelectric Elements [12]

Objective:
• Design and implement a human-footstep-based floor tile using piezoelectric transducers capable of converting kinetic energy from footsteps into electrical power.
• To use inexpensive, accessible materials for application purposes.

Scope:
1. Improved Circuit Design: Optimization of rectifier and storage circuits to increase the conversion efficiency and output power.
3. Integration with IoT: Powering wireless sensors for smart homes, smart cities, and security systems directly.

Limitations
•Low power input: The harvested power currently ≈ 249.6 mW, which is enough to drive only low power devices and not large appliances.
• Durability Issues: Long application of mechanical stress may cause degradation in piezoelectric materials and foam plates.
• High Initial Installation Effort: Large-scale deployment requires careful design, proper flooring integration, and maintenance.

13. Performance and Behavior Analysis of Piezoelectric Energy Harvesting Floor Tiles [13]

Objective:
This work explores two piezoelectric floor tile designs which convert human footsteps into electricity by frequency up-conversion mechanisms: magnetic plucking and mechanical impact. Experimental results showed that the mechanical impact-based tile yielded higher energy output, with a simpler design and higher durability.

Scope:
The proposed study will focus on the subject of vibration energy harvesting using piezoelectric materials to power low-power electronics, such as smart building infrastructure, sensor networks, and public spaces with high pedestrian density.

Limitations:
Magnetic plucking design is structurally complicated and may shorten the life of a piezoelectric element because of direct contact stress.
Energy output is very limited and may not power high-consumption devices. Performance depends greatly on consistent pedestrian activity.

14.Development of an Energy Harvesting Tile Using Novel MXene-Cement Based Triboelectric Nanogenerator [14]

Objective:
By incorporating MXene fillers into cement composites, a triboelectric nanogenerator (TENG) was developed, showing much higher power generation from footsteps than pristine cement-based devices.

Scope:
The research integrates nanomaterials with construction materials for the development of smart building infrastructure that can produce renewable energy. Applications include pavements, walkways, and public spaces.

Limitations:
Integration of MXene introduces cost and complexity during large-scale production.
Long-term durability and environmental stability of MXene-cement composites have not been tested so far.
Output power is modest and suitable only for small-scale electronics. Achievements:
It reached a maximum power output of 134 µW, which is more than twice that of traditional cement-based TENGs.
Harvested human footprint energy successfully: ~10 V peak-to-peak.
Demonstrated the feasibility of multifunctional cements for sustainable infrastructure.

15. Kinetic Energy Harvesting of Footfalls on Specially Designed Floor Tiles [15]

Abstract:
This paper presents an experimental model of converting footstep energy into electrical energy using piezoelectric tiles. A feasibility study at the University of Jordan estimated the potential energy generation if deployed in high-traffic areas.

Scope:
The research targets universities, malls, stations, and public walkways to supplement electricity needs with pedestrian-based energy harvesting. It also explores storage and direct applications like LED lighting.
Limitations:
Energy generation depends a lot on foot traffic volume.
The power produced is small for large-scale electricity. This needs a rectification and storage circuitry, which complicates it.

16. Design and Simulation of Smart Flooring Tiles using Two-Phased Triangular Bimorph Piezoelectric Energy Harvester [16]

Objective:
This paper proposes a triangular bimorph piezoelectric design with stopper mechanisms to harvest energy twice from a single footstep. Simulations using COMSOL demonstrated improved efficiency over rectangular designs.

Scope:
Numerical modeling and optimization of piezoelectric harvesters for urban flooring systems are the focus of the work, targeting transport hubs, markets, and academic institutions.

Limitations:
- Results are mainly simulation-based and require real-world testing for validation.
- Scaling up the fabrication of triangular bimorph structures can be challenging.
- Currently suitable only for low-power applications due to limited energy storage capacity

3. IDENTIFICATION OF THE PROBLEM / SCOPE OF THE PROJECT

The urban areas of today are facing increasing population density, translating into high energy consumption, congestion of roads, and environmental hazards. While progress in technology has been made, much of the existing infrastructure in towns and cities is still heavily reliant on non-renewable energy sources such as coal and fossil fuels. These conventional energy resources contribute to environmental pollution and a surge in operational costs, making sustainable alternatives more crucial than ever.

While most mechanical energy produced by traffic and pedestrians in public areas, like railway stations, airports, shopping malls, or along streets, is dissipated directly as waste heat, current mechanisms to collect and reuse such abundant kinetic energy are inefficient. Ideally, there is a need to develop an efficient approach independently of the geographical and climatic constraints of renewable energy sources like solar and wind, whose efficiencies are very limited.

Moreover, kinetic energy harvesting technologies are not free from various inefficiencies due to the lack of real-time monitoring and intelligent management systems, which create uncertainties in energy generation and increase the cost of maintenance. This calls for an integrated system where energy harvesting is combined with smart analytics to meet performance and reliability considerations.

This project, "AI-Driven Kinetic Tiles for Sustainable Energy Harvesting and Smart Infrastructure," will seek to address these challenges by marrying kinetic energy harvesting with Artificial Intelligence. The project scope entails:

1. Energy Conversion: Designing kinetic tiles which convert mechanical pressures from footfalls or vehicles into usable electrical energy through piezoelectric or electromagnetic components.

2. AI-Enhanced Optimization: Using artificial intelligence to study real-time data on pedestrian movement, environmental conditions, and energy output in a bid to improve generation efficiency and predict maintenance needs.

3. Data Integration and IoT: Using IoT-enabled sensors for continuous data collection and remote monitoring, feeding into AI systems for predictive analytics and smart energy management.

4. Urban Deployment and Sustainability: The tiles will demonstrate practical use in urban environments for charging lighting, signage, and low-energy devices, reducing reliance on conventional power grids.

5. Scalability for Future Smart Cities: The framework should be scalable and adaptive to large infrastructures, such as airports, railway stations, or smart city zones, with a possible integration into hybrid renewable energy systems.

Fundamentally, the work transforms kinetic energy into a green, reusable resource with the use of AI and IoT in the building of efficient, intelligent, and eco-friendly urban infrastructure that aligns with the goals of smart cities.

4. PROPOSED IDEA / SOLUTION

The project "AI-Driven Kinetic Tiles for Sustainable Energy Harvesting and Smart Infrastructure" presents an innovative concept whereby kinetic energy from human footsteps and vehicular movements has been converted to electrical power through intelligent kinetic tiles. The core idea of this development is the union of conventional energy harvesting methods with modern Artificial Intelligence systems that monitor, optimize, and manage the generation of energy in real-time.

The solution involves embedding piezoelectric or electromagnetic transducers within floor tiles that convert mechanical pressure into electrical energy while people or vehicles move across them. Each tile is connected to a microcontroller platform gathering data from sensors tracking energy output, foot traffic intensity, and environmental factors.

Power harvested is stored either in batteries or capacitors for use in low-energy devices, such as LED streetlights, digital signage, or sensors. The integrated AI system analyzes sensor data to optimize energy use, predict maintenance needs, and efficiently distribute power based on real-time demand patterns.

Key capabilities of AI include:

- Predicting energy generation based on modeling pedestrian and vehicular traffic trends.
- Early detection of mechanical wear or system faults for proactive maintenance.
- Power flow management: The ability to control the flow of energy within the plant to ensure that high-demand areas get preference, while at the same time minimize wastage of resources.
- Providing valuable data insights for city planners and infrastructure managers.

These tiles are being implemented along highly trafficked areas like walkways, transit hubs, and campuses. The sensor data is sent over IoT protocols to cloud platforms where AI processes the information for continuous learning and system improvements.

The expected outcomes of such research encompass the transformation of wasted kinetic energy into a clean, renewable resource, enhancing system reliability and efficiency through AI monitoring, reduction of dependence on fossil fuels, and supporting the development of sustainable, smart urban infrastructures.

1. OBJECTIVES OF THE IDEA / SOLUTION AS DESIGN PROJECT

The objective of the project AI-Driven Kinetic Tiles for Sustainable Energy Harvesting and Smart Infrastructure is to focus on the design and development of an intelligent system with the capability to convert kinetic energy from human footsteps and vehicular movement into electrical energy with high efficiency. This will also provide the opportunity to apply Artificial Intelligence technologies, enhancing functionality, performance optimization, and data-driven decision-making.

Key objectives include:

- Efficient Renewable Energy Harvesting: Design kinetic tiles embedded with either piezoelectric or electromagnetic transducers that efficiently convert mechanical pressure into electrical energy.

- AI-driven smart monitoring: Apply machine learning algorithms that continuously analyze real-time sensor data to enable dynamic optimization of energy generation, distribution, and consumption.

- Predictive Maintenance Framework: AI and IoT analytics for wear forecasting and fault prediction, allowing proactive maintenance and bringing down system degradation.

- Intelligent Energy Management: Implement an AI-powered energy management system that prioritizes its flow towards high-demand areas with minimal wastage.

- Real-time Data Acquisition and Analytics: Utilize IoT-connected sensors and cloud-based platforms to permit real-time data acquisition, visualization, and reporting to support infrastructure planning

- Environmental Sustainability Promotion: Support the reduction of reliance on fossil fuels by providing a decentralized, clean, and renewable energy source for lower carbon emissions.

- Demonstrate Practical Urban Use: Validate the technology for real-world applications in public transit stations, airports, educational institutions and commercial complexes to power devices that require low energy consumption.

- Scalable and Cost-Effective Design: Ensure modularity and ease of deployment for broad application with manageable installation and maintenance costs.

- Advocacy for Green Technologies: Increase awareness and acceptance of sustainable energy technologies by showcasing the benefits and applications of kinetic energy harvesting combined with AI analytics.

2. PROCESS / METHODOLOGY

a. Identify the needs of customers and users through design thinking methods

Energy consumption is growing day by day in today's city environments, with a huge dependency on non-renewable sources. Public and commercial areas like shopping malls, railway stations, airports, and educational campuses are some of the places where the movement of human beings is maximum in a day. However, this kinetic energy is wasted without having any mechanism to convert it into useful power.

Through design thinking methodologies, user needs were identified by focusing on three key aspects:

1. Energy Sustainability: The need for renewable energy resources that are efficient, cost-effective, and eco-friendly.
2. Smart Infrastructure: The Need for Intelligent systems able to monitor, optimize, and manage energy resources autonomously.
3. Public Benefit and Awareness: To raise awareness and involve the community in saving energy by means of noticeable, interactive mechanisms like kinetic tiles.

These observations thus made apparent that the development of an AI-driven kinetic energy harvesting system would meet the increasing needs of public space for sustainability and smart energy management.

b. Translate the needs of customer into product / process / project

Based on the identified needs, the project translates those requirements into a tangible solution by:

AI-powered kinetic tiles.

- It converts mechanical energy from human footsteps or vehicular motion into electrical energy through the use of piezoelectric or electromagnetic transducers.
- AI algorithms are integrated to optimize energy production, monitor the health of systems, and predict maintenance needs.

The process involves designing the hardware for energy capture and integrating it with software for data analysis and real-time energy optimization.

- The envisioned project outcome is a self-sustaining, AI-enhanced renewable energy platform suitable for urban environments, contributing to the development of smart cities.

c. Product / process / project specification

Component / Specification / Function

Kinetic Tile Surface

Sensors

Microcontroller Unit

AI Module

Energy Storage

Data Transmission

Output Devices

Piezoelectric material or electromagnetic coil for conversion of mechanical stress into electrical energy.

Pressure and vibration sensors detecting movement and intensity of force.

Acquisition and control system shall be based on Arduino/Raspberry Pi.

Machine Learning for predictive analysis, fault detection, and energy optimization.

Rechargeable lithium-ion battery or capacitor to temporarily store energy.

IoT connectivity for real-time data monitoring and analytics via Wi-Fi or MQTT.

LED lights, display panels, or charging ports that are powered by the energy harvested.

d. Designing the solution for required customer needs

The mechanical, electrical, and AI-driven components come together during the design process in a unified way:

1.Prototype Development:

o Design and fabricate a small-scale model of kinetic tiles utilizing piezoelectric materials.

o Connect sensors and transducers to a microcontroller to collect signals.

2. Integration of AI and IoT:

o Apply AI models that can analyze users' movement data to optimize energy generation patterns.

o Utilize IoT systems for data storage and visualization in the cloud.

3. Testing and Optimization:

- Perform real-time testing in simulated environments to measure efficiency, energy output, and system durability.

-Calibrate the AI model for better prediction accuracy and adaptive control.

4. Final Implementation:

- Deploy the system in high-traffic areas, then observe the performance.

-Collect insights for future scalability and integration into larger smart infrastructure networks

e. Process/ Flow/ Design Diagram

Textual Description of the Flow:

AI-Driven Kinetic Tiles follow a structured workflow that integrates mechanical energy harvesting, sensor data acquisition, AI-based analysis, and smart energy management.

Step-by-Step Flow:

1. Energy Generation-Input Stage

- When a person walks or applies pressure to the kinetic tiles, the piezoelectric sensors transform mechanical pressure into electrical energy.

- The current produced is directed to the energy storage module.

2. Energy Storage and Regulation:

- Electrical energy is stored either in a rechargeable battery or supercapacitor.

-A voltage regulator ensures stable power output suitable for the connected systems.

3. Data Acquisition:

-Sensors: including pressure, motion, voltage, and current that constantly provide step frequency, force intensity, power generated, among other data.

-The collected data then reaches a microcontroller (Arduino / Raspberry Pi) for pre-processing.

4. IoT Layer (Data Transmission):

-Transmission of sensor data using Wi-Fi or Bluetooth modules to a cloud server for monitoring and analysis via ESP8266/ESP32.

5. AI Analysis and Prediction:

-The AI model processes the data to:

• Forecast pedestrian traffic.

-Optimize energy storage and distribution.

-Tile fault or inefficiency detection

-Machine learning algorithms analyze the correlation between movement and energy output.

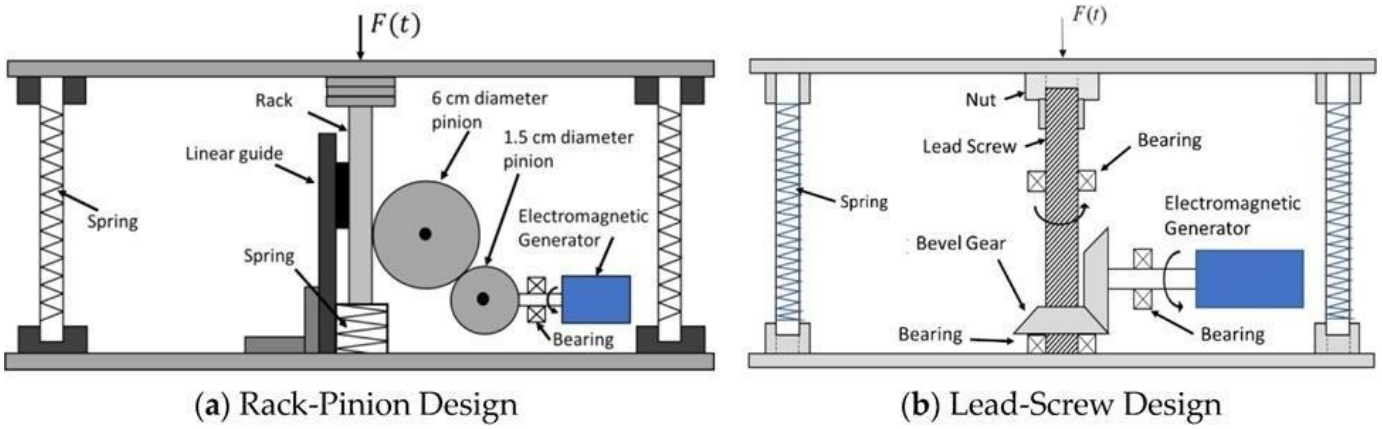
6. Insightful Use and Representation:

-Visualize analyzed data on dashboards using Power BI / ThingSpeak for real-time monitoring. The system adjusts energy usage on its own, depending on the prediction, such as powering LEDs, streetlights, or sensors.

7. Feedback Loop:

-The AI continuously learns from new data, improving prediction accuracy and optimizing system performance over time.

Diagram:



f. Proposed Modules

The whole system of AI-Driven Kinetic Tiles shall be divided into several functional modules, which will all contribute to energy harvesting, monitoring, and intelligent management. Each of the modules have an independent function in the system for high efficiency and scalability.

1. Energy Harvesting Module

This module focuses on capturing the mechanical energy generated from treading or vehicular movement and then converts it into electrical energy by using piezoelectric sensors or electromagnetic induction mechanisms. The harvested energy will be stored in batteries or capacitors for further use.

Functions:

• Capture pressure or vibration input.

• Mechanical stress converted into electrical output.

• Efficiently control and store energy.

2. Sensor and Data Acquisition Module

The information collected in this module includes pressure intensity, energy generated, frequency of tile usage, and other environmental factors in real time. Sensors embedded in the tiles transmit data continuously for analysis and optimization.

Functions:

• Continuous monitoring of tile activity.

• Transmission of data to a central microcontroller.

• Voltage, current, and power output measurement

3. AI Analytics and Optimization Module

The core module performs data analysis through AI algorithms to optimize tile performance. The machine learning models are trained to predict peak usage times, enhance power efficiency, and automatically schedule maintenance.

Functions:

• Predict energy output based on crowd movement patterns.

• Underperforming Tile Identification to Maintenance

Optimize power routing and load distribution.

4. Power Management and Storage Module

This module performs the task of power management, efficiently storing the generated power and supplying it to various connected loads or systems such as streetlights, sensors, and IoT devices.

Functions:

• Power storage in batteries or supercapacitors.

• Real-time regulation of voltage and current.

• Controlled distribution of power to connected systems.

5. IoT Communication and Cloud Integration Module

This module integrates IoT for remote monitoring and control of the whole system. The kinetic tile data gets uploaded to the cloud for real-time analysis and visualization.

Functions:

• Wireless communication using Wi-Fi, LoRa, or MQTT.

• Cloud-based data storage and dashboard visualization.

• Remote system diagnostics and updates.

6. User Interface and Reporting Module

Administrators and users can monitor energy statistics, performance graphs, and maintenance alerts in real time via the web or mobile interface.

Functions:

• Total energy generated and usage trends.

•Provide performance analytics and predictive insights.

Notice users when there are system faults or when maintenance is required.

7.Maintenance and Safety Module

Hardware health, temperature levels, and power surges are monitored by this module to ensure that the system works reliably. It uses AI-based predictive analysis to identify any potential failures before they can occur.

Functions:

• Detect anomalies or overpressure on tiles.

• Predict and prevent system failures.

• Opera with safety and durability.

7.REQUIREMENTS

a. Hardware Requirements

To implement AI-driven kinetic tiles for sustainable energy harvesting and smart infrastructure, hardware components such as the following are necessary:

1.Piezoelectric Sensors / Transducers: These convert mechanical pressure or vibration into electrical energy.

2.Assembly of Electromagnetic Coil and Magnet - Optional energy conversion mechanism for heavy-duty load applications.

3. Microcontroller Unit: The Arduino/Raspberry Pi will be used to collect data from sensors, control the components, and interface with AI modules.

4. Rechargeable Battery / Supercapacitor – to store energy for distribution to output devices.

5. Pressure and motion sensors for the detection of footsteps and the intensity of force applied.

6. Voltage and current sensors are for monitoring energy output and system efficiency.

7. Wi-Fi or Bluetooth Module (ESP8266/ESP32): To enable IoT-based data transmission for real-time monitoring.

8. LED Indicators / Display Units – To visualize system operation and energy utilization.

9. Interconnection Wires, PCB Board, and Enclosure Materials - For assembling the circuit and protection

b. Software Requirements

This software setup is necessary for programming, data analysis, AI model implementation, and performance visualization.

1. Operating System: Windows 10 / Linux / Raspberry Pi OS

2. Languages: Python, C/C++

3. Microcontroller IDE: Arduino IDE / Thonny (for Raspberry Pi)

4. AI & Machine Learning Libraries: TensorFlow, scikit-learn, NumPy, pandas

5. Data Visualization Tools: Matplotlib, Power BI, or Tableau

6.Database / Cloud Storage: Firebase, ThingSpeak, or Google Cloud IoT Core

7. Communication Protocols: MQTT/HTTP for IoT-based data transfer

8. Simulation Software: MATLAB / Proteus / Tinkercad-for circuit design and validation

c. Datasets, if any

Key datasets to be used for AI-driven predictive and analytics modules will be gathered through real-time sensor readings from kinetic tiles, including:

1. Foot Traffic Data: Steps taken, frequency, and impact intensity over time.

2. Energy Generation Data: Voltage and current output corresponding to different load conditions.

3. Environmental Data: Temperature, humidity, and external vibrations that impact system efficiency.

4. Maintenance Data: Tile health, durability patterns, and predictive fault detection.

Additional training and testing of the AI model for predictive energy generation and optimization can be based on synthetic or open-source datasets of human movement patterns.

d. Technologies and Tools

To make the system design, integration, and testing efficient, the following technologies and tools are used:

1. AI/ML: TensorFlow, scikit-learn for predictive analysis and system optimization.
2. IoT Integration: ESP8266 / ESP32 modules for cloud connectivity and data monitoring.
3. Data Storage & Visualization: Firebase, ThingSpeak, or Google Cloud IoT for data handling and visualization dashboards.
4. Circuit Design & Simulation: Proteus, Tinkercad or Fritzing for hardware circuit design.
5. Programming tools: Arduino IDE, Python (Jupyter Notebook or VS Code).
6. Renewable Energy Components: piezoelectric or electromagnetic energy converters for energy harvesting.

8. **NOVELTY OF THE IDEA / PROJECT**

This design introduces an advanced mix of AI, IoT, and kinetic energy harvesting technologies. Its novelty lies in real-time AI-driven optimization of energy generation and predictive maintenance, improving efficiency and reliability over anything seen thus far with traditional systems.

Key Novel Features

1. AI-Driven Energy Harvesting Optimization: The implementation of this project incorporates advanced Artificial Intelligence algorithms that continually study the patterns of pedestrian and vehicular traffic. Through the learning from these real-time data streams, it dynamically adjusts the energy-conversion parameters of the kinetic tiles so as to maximize power generation efficiency across varying conditions of usage and peak times.

2. Predictive Maintenance with AI and IoT: By leveraging sensor data analytics, the tiles will continuously monitor their own operational health by identifying early signs of mechanical degradation or faults. This predictive insight offers the chance for pre-emptive scheduling of maintenance at reduced levels of down time and improved maintenance costs as opposed to conventional reactive upkeep.

3. Smart Energy Management System (EMS): An AI-based EMS intelligently distributes the harvested energy to zones with higher immediate demand, thus minimizing wastage. This targeted allocation further enhances overall system effectiveness, making the kinetic tiles suitable to power real-world applications such as streetlights, information displays, and IoT sensors reliably.

4. Data-Driven Urban Insights and Planning: Beyond energy production, the system collects and aggregates data on pedestrian flows, environmental conditions, and infrastructure usage. This actionable intelligence makes sure urban planners and facility managers to optimize layout designs, crowd control, and resource management in smart city environments.

5. Scalability and Modularity: The tiles can be scaled from small installations in buildings and campuses to expansive urban infrastructures like transit hubs and commercial districts, because the platform is modular. This flexibility accommodates diverse deployment scenarios while enabling integration with hybrid renewable solutions including solar and wind.

6. Sustainability and Environmental Impact: By harnessing normally wasted kinetic energy into clean electricity, the project directly reduces reliance on fossil fuels. In its design lies the facilitation of decentralized power generation, reduced carbon footprints, and a positive contribution to active environmental conservation efforts.

7. Extending User Experience and Interaction: Ability to embed interactivity through third-party applications and enable users to visualize their energy contributions or develop awareness about sustainable best practices. This behavioral feedback loop enhances public participation in energy conservation initiatives.

9.CONCLUSION

The project makes effective use of AI and IoT to convert previously wasted kinetic energy from pedestrian and vehicle movement into renewable electrical power. It provides a practical and scalable solution for modern smart cities, aiming to achieve a balance between energy efficiency, environmental sustainability, and intelligent infrastructure.

Through features like predictive maintenance and real-time energy management, the system enhances the lifespan and performance of urban energy-harvesting installations. It also supports broader environmental goals by reducing dependence on fossil fuels and promoting greener, data-driven city development.

Additionally, the data collected by the system offers valuable insights into human movement patterns and environmental conditions. This information can help urban planners make better decisions in designing smarter and more efficient public spaces. By combining energy generation with intelligent data management, the tiles go beyond being simple power sources—they become key components of a connected, smart city ecosystem.

The design is modular and scalable, making it suitable for a wide range of locations, including campuses, shopping areas, and transport hubs. This versatility expands its impact on energy conservation and sustainability efforts.

By harnessing clean energy and enabling intelligent infrastructure control, the AI-Driven Kinetic Tiles project contributes significantly to building greener, smarter cities. With continued research and development, the system could further improve adaptability, integrate hybrid renewable technologies, and encourage community participation—strengthening its role in the future of sustainable urban living.

In essence, AI-Driven Kinetic Tiles represent a forward-looking step in uniting innovation, sustainability, and technology to create a cleaner and smarter world.

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